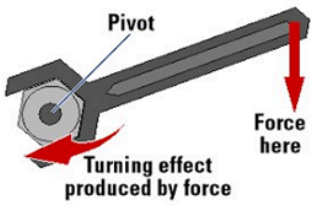
Chapter 4

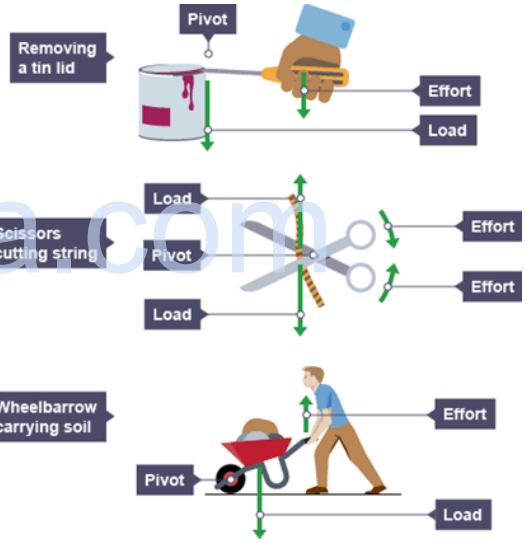
Turning Effects





Describing and Calculating the Turning Force

A turning effect or moment of a force is a measure of how effectively a force can make an object rotate or turn.



Pivot or Fulcrum: The fixed point about which a lever turns.

Moment: Turning effect of a force about a pivot.

- The moment of a force is calculated by multiplying the force applied by the perpendicular distance from the pivot point to the line of action of the force.

- Mathematically, the moment (M) of a force (F) is expressed as **M = F × d**, where d is the perpendicular distance.

Moments are usually **measured in Newton meters (Nm).**
Turning effects have various real-life uses:

Tools and Machines: Turning effects make tools like wrenches and gears in bikes easier to use.
Building and Stability: Architects use them to build strong structures like bridges, and cranes use them to lift heavy things safely.

Principle of Moments and Conditions for Equilibrium

The principle of moments states that an object is in equilibrium when the total clockwise moment about a pivot is equal to the total anticlockwise moment about the same pivot.

- This principle is derived from the concept of rotational equilibrium, where an object remains at rest or in a state of uniform rotation.
- Moment calculations are crucial for determining the balance or stability of objects.

www.edvnova.com

Conditions for Equilibrium

An object is in equilibrium when two conditions are met:

- ★ The forces acting on it are balanced, meaning there is **no resultant force**. This ensures translational equilibrium.
- ★ The turning effects of the forces are balanced, meaning there is **no resultant turning effect.** This ensures rotational equilibrium.

Describing How the Center of Gravity Affects Stability

The center of gravity (CG) is the point within an object where its weight can be considered to act as if all the weight is concentrated at that point.

- An object is stable when its center of gravity is positioned over its base of support. This prevents it from toppling over.

The higher the center of gravity, the less stable the object becomes. Applying the Principle of Moments with Multiple Moments

www.edvnova.com

- ★ In cases where there are multiple forces acting on an object at different distances from a pivot, the principle of moments still applies.
- ★ Calculate the moment for each force individually and then sum the moments to determine if the object is in equilibrium.

Summary:

- The moment of a force is a measure of its turning effect, calculated as the product of the force and the perpendicular distance from the pivot point.
- Mathematically, Moment (M) = Force (F) × Perpendicular Distance (d).
- An object is in equilibrium when the forces on it are balanced (no resultant force), and the turning effects of the forces on it are balanced (no resultant turning effect).
- The center of gravity is the point at which the weight of an object appears to be concentrated. The location of the center of gravity significantly affects an object's stability; a lower center of gravity leads to increased stability.

Questions and Answers

1. What is the moment of a force, and how is it calculated?

The moment of a force is a measure of its turning effect. It is calculated by multiplying the force (F) applied by the perpendicular distance (d) from the pivot point to the line of action of the force. Mathematically, Moment (M) = Force (F) × Perpendicular Distance (d).

2. Explain the principle of moments and how it relates to equilibrium.

The principle of moments states that an object is in equilibrium when the total clockwise moment about a pivot equals the total anticlockwise moment about the same pivot. This principle is crucial for maintaining both translational equilibrium (resultant force is zero) and rotational equilibrium (resultant turning effect is zero).

3. Define the center of gravity and discuss its role in an object's stability.

www.edvnova.com

The center of gravity (CG) is the point within an object where its weight appears to be concentrated. The location of the CG significantly affects an object's stability. When the CG is over the base of support, the object is stable; if the CG is high, stability is reduced.

4. How does increasing the force or the distance from the pivot affect the moment of a force?

Increasing the force or the distance from the pivot both increase the moment of a force. The moment (M) is directly proportional to the force (F) and the perpendicular distance (d): M = F × d.

5. Provide an example of a real-life application of the principle of moments in everyday objects.

A seesaw is an example of the principle of moments in action. It remains level when the moments on one side, caused by the weights of the children, equal the moments on the other side, resulting in rotational equilibrium.

6. How does the choice of support affect the stability of an object?

The choice of support affects an object's stability. A wider base of support increases stability by reducing the chance of toppling. Narrower bases reduce stability.

7. Explain how a tightrope walker uses the principle of moments to maintain balance.

Tightrope walkers use long poles to lower their center of gravity, thereby increasing their stability on the rope. The moment created by their body weight is balanced by the moment created by the pole.

8. What is translational equilibrium, and what conditions must be met for it to occur?

Translational equilibrium means that the resultant force acting on an object is zero. To achieve this, all the forces on an object must cancel each other out in terms of magnitude and direction.

9. Describe an experiment to determine the center of gravity of an irregularly shaped object.

You can determine the center of gravity of an irregular object by suspending it from multiple points and finding the point at which it balances. The intersection of the lines drawn from the suspension points will mark the CG.

www.edvnova.com

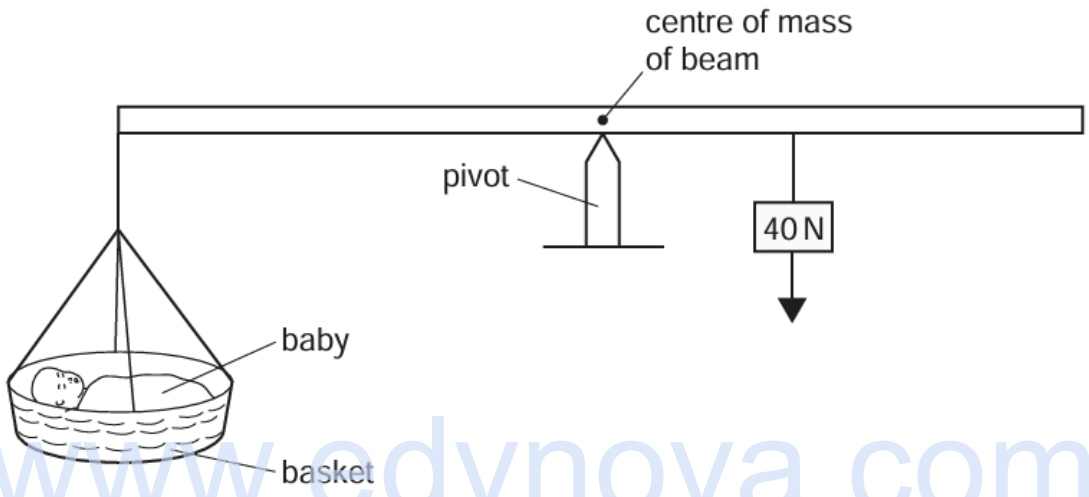
10. How does the stability of an object like a bookshelf change if heavy books are placed on the top shelf compared to the bottom shelf?

Placing heavy books on the top shelf raises the center of gravity of the bookshelf, making it less stable. The increased distance from the base of support increases the likelihood of the bookshelf tipping over.

Past Paper questions

1. The diagram shows a balance being used to find the weight of a baby. The weight of the basket can be ignored.

At equilibrium, the pivot is nearer to the 40 N balancing weight than to the baby.



www.edvnova.com

What is the weight of the baby?

A less than 40 N

B 40 N

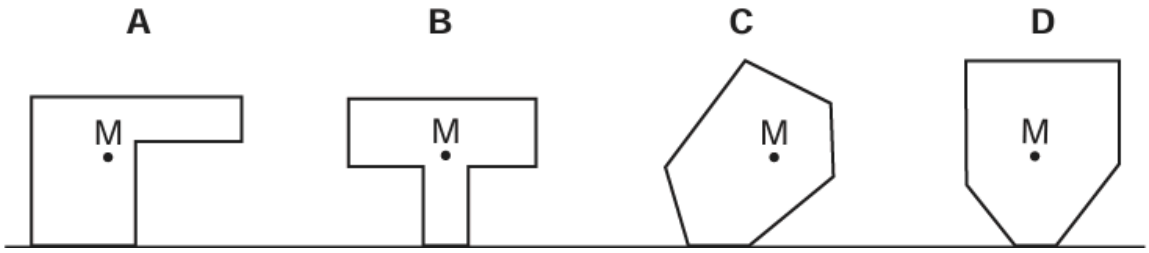
C more than 40 N

D impossible to tell without a scale on the beam

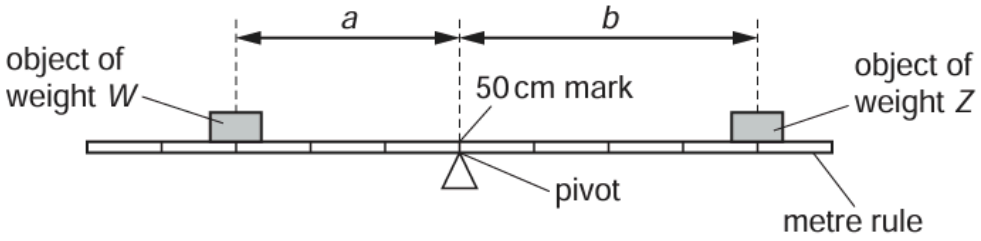
2. The diagram shows four objects on a flat surface.

The centre of mass of each object is marked M.

Which object is about to fall over?



3. The diagram shows an object of weight W and an object of weight Z balanced on a uniform metre rule.



Which equation relating to W, Z, a and b is correct?

A $\frac{W}{a} \quad \frac{Z}{b}$

B $W \times Z = a \times b$

C $W \times a = Z \times b$

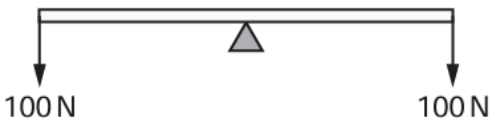
D $W \times (a + b) = Z$

www.edvnova.com

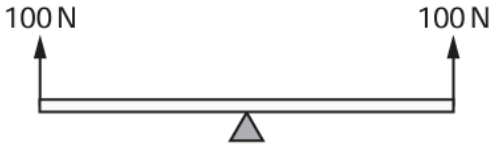
4. A uniform rod rests on a pivot at its centre. The rod is not attached to the pivot. Forces are then applied to the rod in four different ways, as shown. The weight of the rod can be ignored.

Which diagram shows the rod in equilibrium?

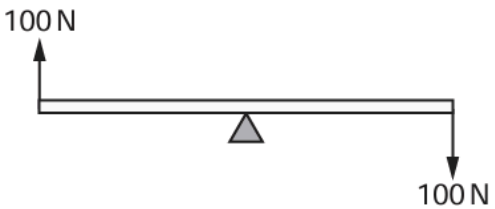
A



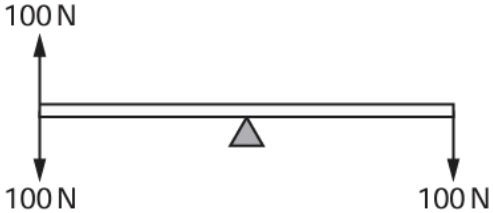
B



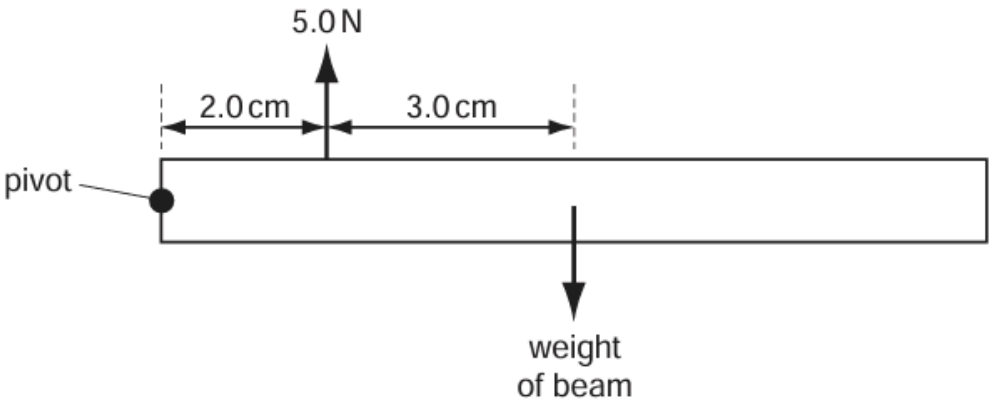
C



D



5. A beam pivoted at one end has a force of 5.0 N acting vertically upwards on it as shown. The beam is in equilibrium.

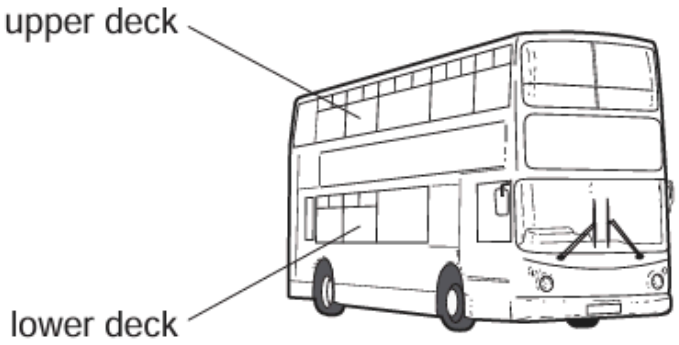


What is the weight of the beam?

A 2.0 N **B** 3.0 N **C** 3.3 N **D** 5.0 N

www.edvnova.com

6. Passengers are **not** allowed to stand on the upper deck of double-decker buses.



Why is this?

A They would cause the bus to become less stable.

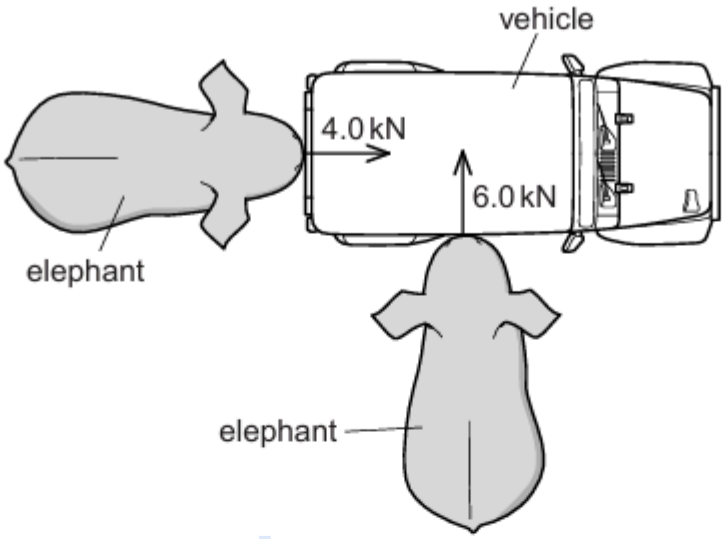
B They would cause the bus to slow down.

C They would increase the kinetic energy of the bus.

D They would lower the centre of mass of the bus.

7.

(a) Fig. 4.1 shows a top view of a tourist vehicle in a game park and two elephants pushing against the vehicle. The two forces indicated are at right angles to each other.



www.edvnova.com

Fig. 4.1

In the space below, draw a scale vector diagram to determine the magnitude of the resultant force. Label the two forces applied and the resultant, and clearly state the scale you use.

magnitude of resultant force = ...[3]

(b) Fig. 4.2 shows another elephant pushing horizontally against a vehicle with a force of 11 kN at a distance 1.8 m above the ground. Point M is the centre of mass of the vehicle.

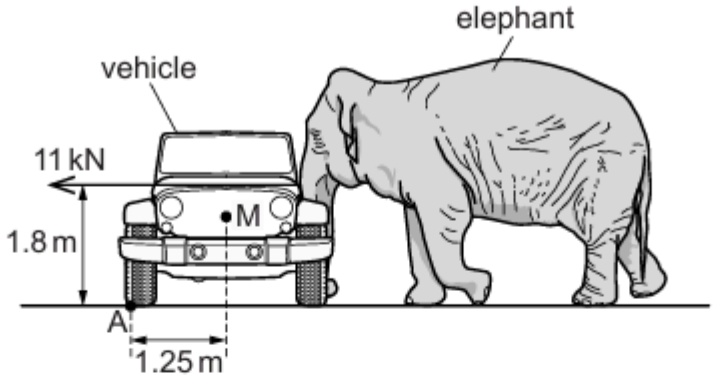


Fig. 4.2

(i) Calculate the moment about point A of the force exerted by the elephant.

www.edvnova.com

moment = ...[2]

(ii) The mass of the vehicle is 1900 kg, and it does not slide when pushed by the elephant.

Determine whether the elephant tips the vehicle over. Show your working.

calculation

conclusion .....[2]

[Total: 7]

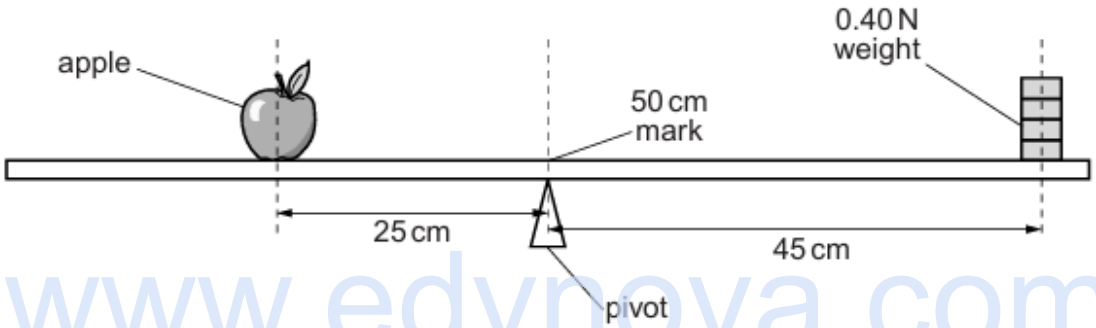
8. A metre rule balances when the 50 cm mark is directly above a pivot.

(a) State where in the rule its centre of mass is located.

.....

..... [1]

(b) Fig. 3.1 shows an apple and a 0.40 N weight placed on the rule so that the rule remains balanced at the 50 cm mark.



www.edvnova.com

Fig. 3.1 (not to scale)

The centre of mass of the apple is 25 cm from the pivot and the centre of mass of the weight is 45 cm from the pivot.

Calculate

(i) the weight of the apple,

weight = ... [2]

(ii) the mass of the apple.

mass = ... [1]

(c) The apple is not moved. The weight is removed from the rule and the pivot is moved to the left until the rule balances as shown in Fig. 3.2.

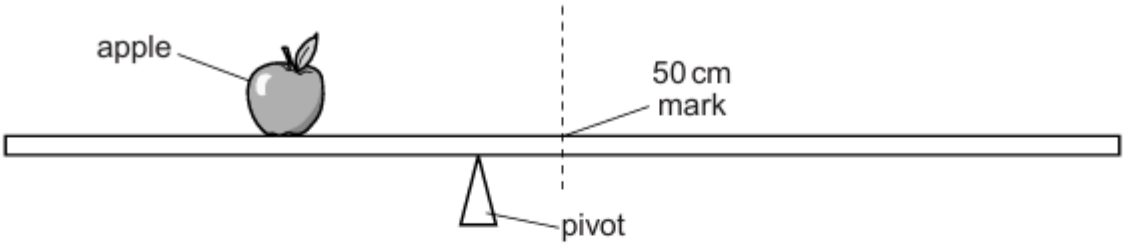


Fig. 3.2 (not to scale)

(i) Explain why the arrangement in Fig. 3.2 balances.

.....

.....

..... [2]

www.edvnova.com

(ii) The pivot in Fig. 3.2 is closer to the 50 cm mark than to the centre of mass of the apple.

Compare the weight of the rule to the weight of the apple.

.....

..... [1]

[Total: 7]

10.

(a) **(i)** Write down the names of **three** man-made devices in everyday use that depend, for their action, upon the moments of forces.

1. .....

2. .....

3. .....

[2]

(ii) Fig. 3.1 shows a uniform rod AB acted upon by three equal forces *F*.

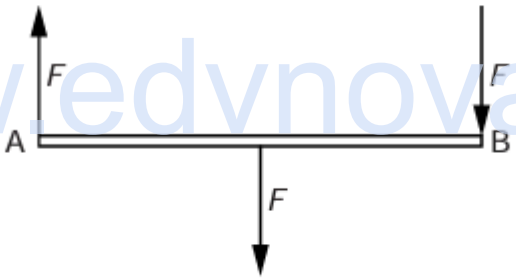


Fig. 3.1

State **two** reasons why the rod is **not** in equilibrium.

1. .....

2. .....

[2]

www.edvnova.com

(b) Fig. 3.2 shows a uniform rod PQ, supported at its centre and held in a horizontal position. The length of PQ is 1.00 m.

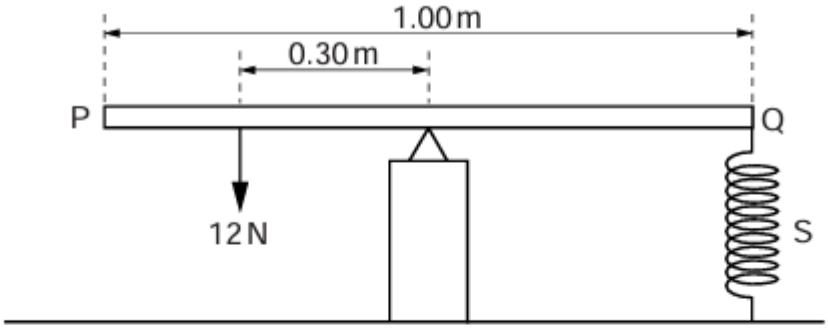


Fig. 3.2

www.edvnova.com

A force of 12 N acts at a distance of 0.30 m from the support. A spring S, fixed at its lower end, is attached to the rod at Q.

(i) Calculate the force exerted on PQ by the spring.

force = ... [2]

(ii) Explain why it is not necessary to know the weight of PQ.

.....

..... [1]

[Total: 7]

www.edvnova.com